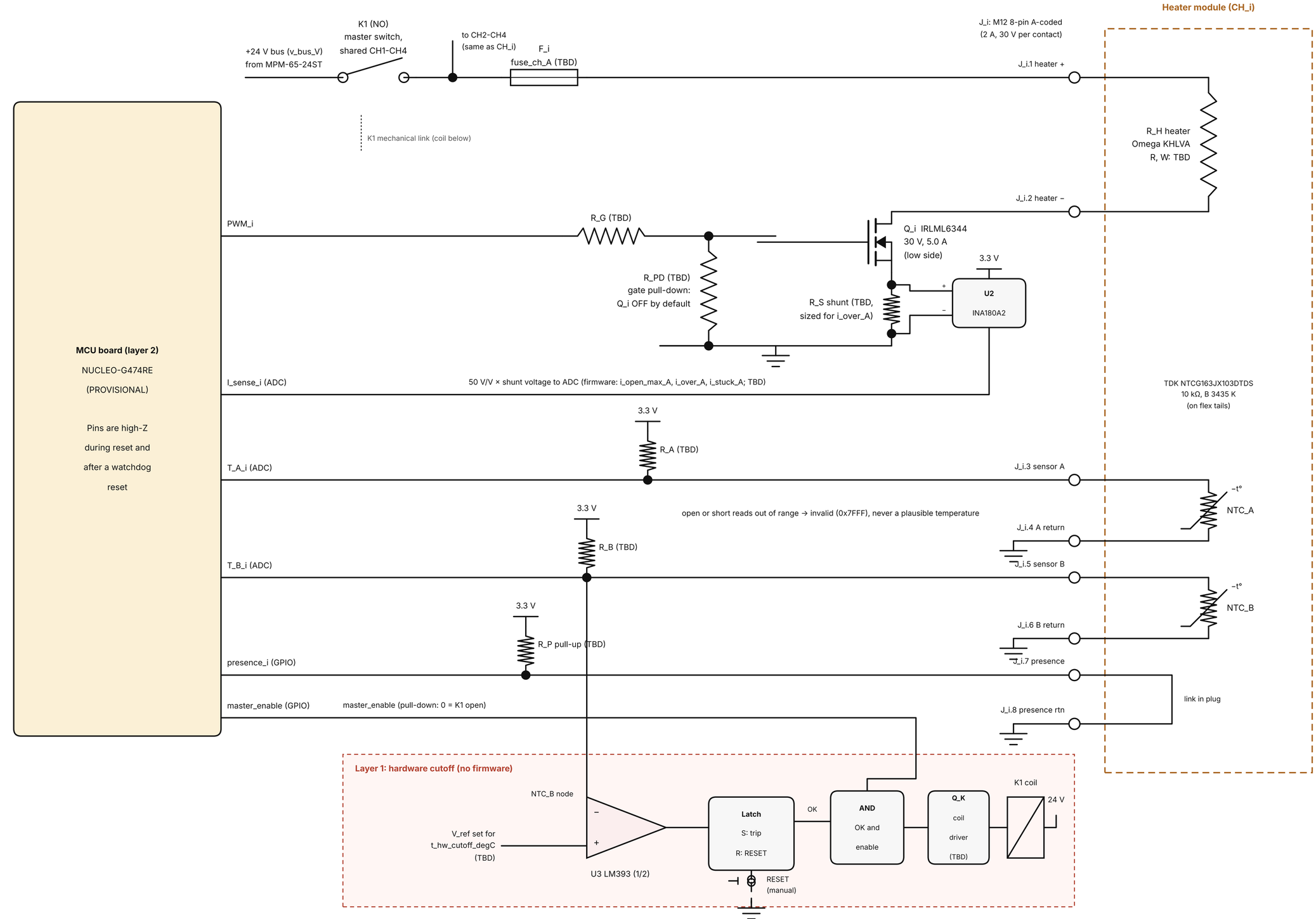
NERAM one-channel schematic: heater channel CH_i with master switch, current sense and layer-1 hardware cutoff

PROVISIONAL topology (Sai draft for Shreehan, 18 Oct 2026). Paper design: not built, not simulated, not tested. Parts from bom/electrical_bom.csv; every value marked TBD is unsourced and must not be guessed.



Default-off (all must hold with no firmware)

- K1 is normally open: no coil current → no heater power
- master_enable has a pull-down; MCU pins are high-Z in reset
- R_PD holds Q_i off when PWM_i floats
- A cutoff trip latches until the manual RESET

Who can remove heater power

- Layer 1: comparator → latch → K1 opens
- Layer 2: MCU sets PWM_i = 0 and master_enable = 0;
  watchdog reset → pins high-Z → same result
- Layer 3: controller sends Heat_i = 0 (via the MCU)

Stuck-on driver (Q_i shorted)

- current while commanded off → firmware opens K1;
  over-temperature → layer-1 cutoff opens K1

Open items (owner Shreehan)

- All values marked TBD; shunt and V_ref from i_over_A
  and t_hw_cutoff_degC once those are sourced
- The 8 contacts are all used (heater 2, A 2, B 2,
  presence 2), so the cutoff reads the NTC_B divider;
  a buffer may be needed so the ADC and the
  comparator do not load each other
- Q_i: 30 V rating on a 24 V bus (check transients)
- Flyback/snubber for K1 coil; fuse rating
- Output state of the Nucleo pins at reset: verify

Proof-of-concept prototype, not a medical device. No hardware has been built. Standards are design references only; no compliance is claimed.